# On the Conditioning of Diagonal State Space Models Under Quantization-Aware Training

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(Dated: June 2, 2026)

Diagonal state space models (S4D) admit many initializations, and these differ by twelve orders of magnitude in the conditioning of their eigenvector matrix $\kappa(V)$—from $\kappa(V) \approx 14$ for the Fourier-aligned FouT (S4D-Lin) to $\kappa(V) \approx 10^{13}$ for the HiPPO-derived Skew-HiPPO. Because $\kappa(V)$ measures how strongly small parameter perturbations are amplified, classical perturbation theory (the Bauer–Fike theorem) predicts that quantizing the ill-conditioned initialization should corrupt its convolution kernel some $10^{12}\times$ more than the well-conditioned one. We test this prediction on sCIFAR-1024 across six bit-widths and three seeds. It does not survive per-channel quantization-aware training (QAT): the empirical kernel-error ratio collapses from $10^{12}\times$ to 3–4$\times$, and the FouT/Skew-HiPPO test-accuracy gap holds at a roughly constant $\sim$3 pp even at full precision—so it reflects a baseline difference between the initializations rather than their quantization sensitivity. A frozen-$\Delta$ ablation rules out timestep drift as the mechanism of conditioning erosion: drift in $A_{\text{real}}$ and $A_{\text{imag}}$ alone pushes $\kappa$ to HiPPO magnitude within six epochs. These findings invite a re-reading of HiPPO theory: differently-initialized diagonal SSMs converge to the same conditioning regime despite starting twelve orders of magnitude apart, which suggests that the HiPPO-optimal operating point is a *destination* property of diagonal-SSM training rather than a property of the parameterization—initialization sets the optimization trajectory, not its high-$\kappa$ endpoint.

## I. INTRODUCTION

State space models (SSMs) [2, 3] have emerged as a competitive alternative to attention-based architectures for long-sequence modeling. The diagonal parameterization (S4D) is particularly attractive for deployment: its Vandermonde-structured convolution kernel is efficient to compute, and its small parameter count makes it a natural target for quantization-aware training (QAT). Recent work has begun to apply QAT and post-training quantization to SSMs and Mamba variants [9–12], generally finding that per-channel (per-mode) quantization is required for acceptable accuracy at low bit-widths. The theoretical reason for this engineering convergence has not been characterized.

This paper investigates how initialization-induced conditioning of the diagonal SSM eigenvector matrix translates to task-level robustness under quantization-aware training. The Bauer–Fike theorem [5] links perturbations of the diagonal parameters to kernel error via the eigenvector condition number $\kappa(V)$. The S4D catalog of initialization choices spans twelve orders of magnitude in $\kappa(V)$: the HiPPO-derived Skew-HiPPO [1] sits at $\kappa(V) \approx 10^{13}$ via its highly non-normal state matrix, while the Fourier-aligned FouT (S4D-Lin [3])—which is not derived from polynomial projection against a HiPPO measure—sits at $\kappa(V) \approx 14$ via uniform eigenvalue placement on the unit disk. We characterize how this conditioning spread plays out at the task level on sCIFAR-1024, across bit-widths and random seeds, and use an ablation freezing the timestep parameter to identify which SSM parameters drive the conditioning trajectory during training. Our aim is to determine whether the Bauer–Fike prediction translates from kernel-error sensitivity to task accuracy, and to identify the design choices that govern the outcome.

## II. THEORY

### A. HiPPO theory: signal compression as a quadrature problem

HiPPO (High-order Polynomial Projection Operator) [1] formulates online signal compression as a *quadrature* problem. Given a continuous input signal, the goal is to maintain at each time $t$ a finite-dimensional summary of the past that supports fast online updates and admits accurate reconstruction. The underlying assumption is that the input signal lives on a continuous manifold and can be approximated by polynomial expansion against a time-varying measure.

Formally, let $\mu(t)$ be a time-varying measure family supported on $(-\infty, t]$, let $G$ be an $N$-dimensional subspace of polynomials, and let $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ be continuous. HiPPO defines, at every time $t$:

- A **projection operator** $\text{proj}_t$ that takes the input history $f|_{\leq t}$ and maps it to the polynomial $g_t \in G$ that best approximates it in $L^2(\mu(t))$.
- A **coefficient operator** $\text{coef}_t$ that maps the polynomial approximation $g_t$ to its coordinates $c(t) \in \mathbb{R}^N$ in the orthonormal basis of $G$ defined with respect to $\mu(t)$.

The coefficient vector $c(t)$ evolves according to a linear ODE:

$$\frac{dc}{dt} = A(t)\, c(t) + B(t)\, f(t), \qquad c(t) \in \mathbb{R}^N, \quad (1)$$

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where $A(t), B(t)$ are determined entirely by the choice of $\mu(t)$. The reconstruction error obeys

$$\big\|f|_{\le t} \;-\; g_t\big\|^2_{L^2(\mu(t))} \;\longrightarrow\; 0 \qquad \text{as } N \to \infty, \tag{2}$$

with convergence rate depending on the regularity of $f$ and the measure. The most common choice—the scaled Legendre measure on $(0, t]$, giving HiPPO-LegS—admits tractable closed-form expressions for $A$ and $B$.

*a. What the measure buys.* The scaled Legendre measure—uniform on $[0, t]$ with normalizing factor $1/t$—is what gives HiPPO-LegS its defining property. Under this measure, the shifted-and-scaled Legendre polynomials $\{P_n^{(t)}\}_{n=0}^{N-1}$ form an orthonormal basis of $G$, and standard orthogonal-polynomial approximation theory guarantees that $\mathrm{proj}_t(f|_{\le t})$ is the best $L^2(\mu(t))$ approximation of the input history among polynomials of degree $< N$ [4].

*b. From projection to kernel: where perturbations enter.* The projection ODE (1) is implemented in practice by the SSM convolution kernel introduced in Section II B below. Computing the discrete evolution of $c(t)$ efficiently requires diagonalizing $A$ and convolving with the resulting Vandermonde-structured kernel; any perturbation to the diagonal parameters thus propagates into the kernel and, downstream, corrupts the polynomial approximation underlying (2). The **Bauer–Fike theorem** [5] quantifies this propagation: for a diagonalizable matrix $A = V\Lambda V^{-1}$, any perturbation $A + E$ produces eigenvalues contained in discs of radius $\|E\| \cdot \kappa(V)$ around the original spectrum, where the *eigenvector condition number*

$$\kappa(V) \;=\; \|V\|\,\|V^{-1}\| \tag{3}$$

is the amplification factor. Through the Vandermonde structure of the SSM kernel, this in turn yields kernel error scaling with $\|E\| \cdot \kappa(V)$ (Appendix A gives the formal bound). The condition number $\kappa(V)$ thus measures how robust the HiPPO approximation is to perturbations of the diagonal parameters.

*c. The structural source of large $\kappa(V)$.* The HiPPO-LegS state matrix is highly **non-normal** ($AA^* \ne A^*A$), so its eigenvectors are not orthogonal under the standard Euclidean inner product, and $\kappa(V)$ grows rapidly with $N$. Empirically $\kappa(V) \approx 1.3 \times 10^{13}$ at $N = 64$ for the Skew-HiPPO normal component used by S4D, making the diagonalization of $A$ a numerically **ill-posed problem** [7]. The non-normality is structural rather than incidental: the LegS dynamics encode the action of a translating, rescaling integration domain on Legendre coefficients, which is intrinsically asymmetric and resists orthogonal diagonalization. The richness of the polynomial-projection structure *is* the conditioning problem.

*d. Initialization as a continuous tradeoff.* The catalog of S4D initializations—LegS, Inv, Lin, Real—is typically presented as a discrete menu of choices benchmarked against task accuracy. We argue that the catalog is better understood as a continuous axis: the $\kappa(V)$-versus-memory-structure tradeoff. HiPPO theory privileges the high-$\kappa$ end (rich polynomial-projection structure, strong memory guarantees); numerical optimization and quantization robustness privilege the low-$\kappa$ end (well-conditioned eigenbasis, small perturbation amplification). Yu et al. [8] characterize a corresponding approximation-estimation tradeoff in SSM optimization theory. At $N = 64$ and $\Delta = 2/N$, $\kappa(V)$ ranges from $\approx 14$ for FouT to $\approx 1.3 \times 10^{13}$ for Skew-HiPPO, with S4D-Inv and S4D-Real falling at intermediate values. We examine the conditioning side of this tradeoff under quantization-aware training; Section IV reports how the predicted ranking plays out against actual kernel error, task accuracy, and conditioning trajectory over training.

### B. From the HiPPO ODE to diagonal SSMs

The continuous-time SSM corresponding to (1) is

$$\dot{x}(t) \;=\; A\,x(t) + B\,u(t), \qquad y(t) \;=\; C\,x(t) + D\,u(t), \tag{4}$$

which discretizes (e.g., via zero-order hold) to $x_k = \bar{A}\,x_{k-1} + \bar{B}\,u_k$, $y_k = \bar{C}\,x_k + \bar{D}\,u_k$ with $\bar{A} = e^{A\Delta}$. The length-$L$ convolution kernel is

$$K_\ell = \bar{C}\,\bar{A}^\ell\,\bar{B}, \qquad \ell = 0, \ldots, L-1. \tag{5}$$

The original S4 parameterization [2] decomposes $A$ as a normal matrix plus a rank-1 perturbation (DPLR), which requires careful handling of the low-rank component. **S4D** [3] simplifies by parameterizing $A$ as fully diagonal in $\mathbb{C}$:

$$A = \mathrm{diag}(\lambda_1, \ldots, \lambda_N), \qquad \lambda_n \in \mathbb{C}_-, \tag{6}$$

where $\mathbb{C}_-$ denotes the open left half-plane (for stability). The discretized eigenvalues are $\bar{\lambda}_n = e^{\lambda_n \Delta}$, and the convolution kernel becomes a Vandermonde product:

$$K_\ell = \sum_{n=1}^{N} \bar{c}_n\,\bar{b}_n\,\bar{\lambda}_n^\ell, \qquad V_{\ell,n} = \bar{\lambda}_n^\ell. \tag{7}$$

The matrix $V \in \mathbb{C}^{L\times N}$ is the **discrete-time Vandermonde** induced by the diagonal eigenvalues and timestep. Its condition number $\kappa(V)$ is the quantity that appears in (3) and that governs the kernel's sensitivity to perturbations of the diagonal parameters (including, in particular, those introduced by quantization).

One property of this parameterization is that $\kappa(V)$ depends on *both* the eigenvalue layout $\{\lambda_n\}$ and the timestep $\Delta$ through $\bar{\lambda}_n$; an initialization that is well-conditioned at one $\Delta$ can be catastrophically ill-conditioned at another. Another is, $\bar{\lambda}_n$, $\bar{c}_n$, $\bar{b}_n$ are all derived from learnable parameters, so $\kappa(V)$ is a function of the current training state, not a fixed property of the initialization.

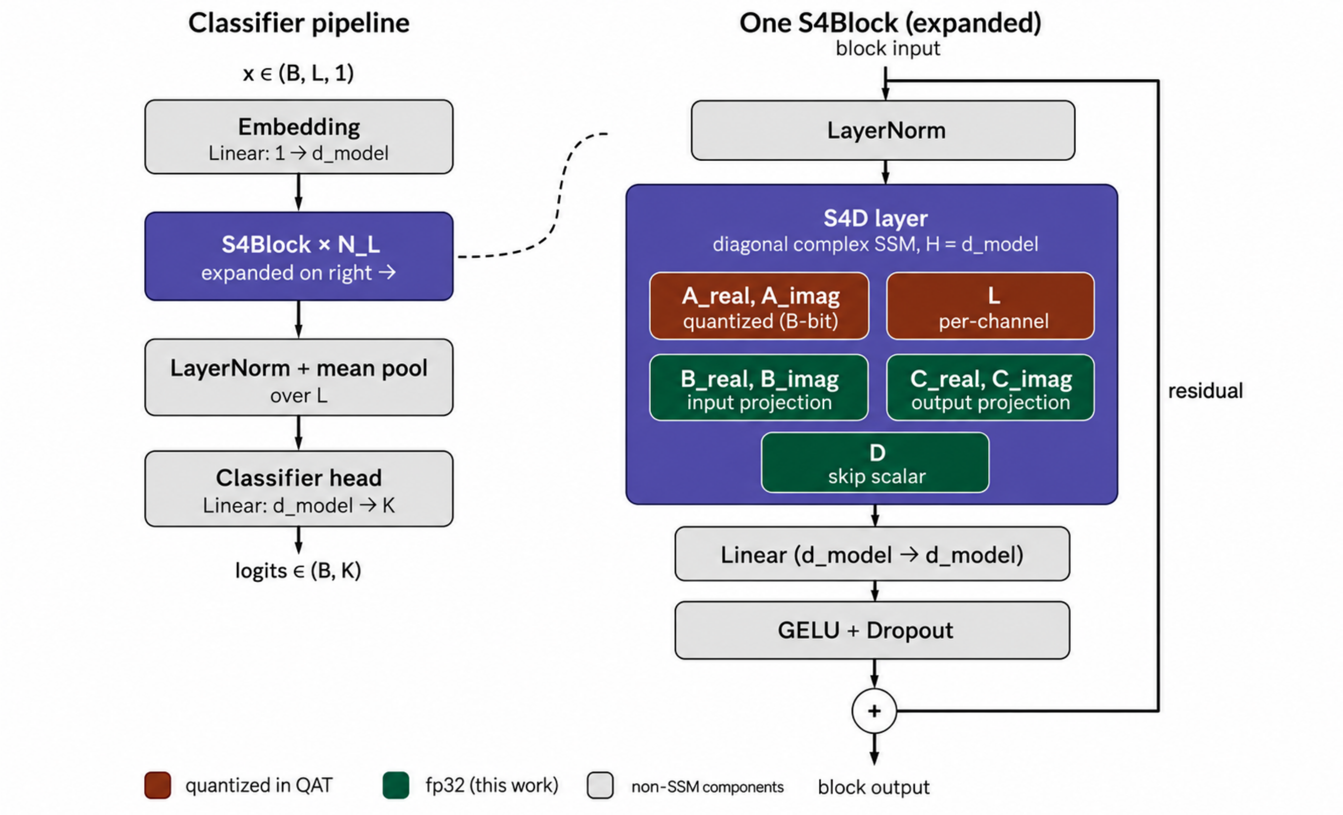


FIG. 1. S4D classifier architecture. *Left:* macro pipeline. *Right:* one S4Block expanded, with S4D layer parameters partitioned by quantization treatment. Coral parameters ($A_{\text{real}}, A_{\text{imag}}, \log\Delta$) are quantized under QAT; teal parameters ($B, C, D$) remain fp32 in this work.

## III. METHOD

### A. Initialization

We compare two initializations on the $\kappa(V)$ axis (Section II A):

- **FouT (S4D-Lin):** $\omega_n = \pi n$, $\alpha_n = \frac{1}{2}$ for $n = 1, \dots, N/2$. Eigenvectors of the discrete-time Vandermonde are Fourier modes; $\kappa(V) \approx 14$ at $\Delta = 2/N$.

- **Skew-HiPPO (S4D-LegS):** $\omega_n$ are the imaginary parts of the eigenvalues of $A_{\text{normal}} = A_{\text{LegS}} + \frac{1}{2}PP^\top$ (the normal component of the LegS DPLR decomposition); $\alpha_n = \frac{1}{2}$. $\kappa(V) \approx 1.3 \times 10^{13}$ at $\Delta = 2/N$.

The 12-order-of-magnitude $\kappa$ gap at initialization is the empirical anchor for the Bauer–Fike prediction; we report results on S4D-Inv and S4D-Real (intermediate $\kappa$) in the appendix.

*a. Timestep initialization.* We initialize $\log\Delta_h$ at the Nyquist scale with small per-channel jitter:

$$\log\Delta_h = \log(2/N) + \epsilon_h, \qquad \epsilon_h \sim \mathcal{N}(0, \sigma_\Delta^2), \tag{8}$$

with $\sigma_\Delta = 0.1$. This centers all channels in the $\kappa$-minimizing band for FouT (Phase 1 measurements in Section C show $\kappa(V)$ grows by $\sim 10^{11}$ within 10% jitter off Nyquist). As ablation, we also report results with the standard log-uniform initialization $\log\Delta \sim U[\log 10^{-3}, \log 10^{-1}]$.

### B. Quantizer

We use $B$-bit symmetric uniform quantization with quantization step $\Delta_Q = s/(2^{B-1} - 1)$:

$$Q_B(x) = \Delta_Q \cdot \text{round}\left(\frac{x}{\Delta_Q}\right), \tag{9}$$

where $s$ is a per-tensor or per-channel scale. Gradients pass through via a straight-through estimator: $\partial Q_B/\partial x := \mathbf{1}[|x| \le s]$.

*a. Per-channel scales.* For per-channel mode, scales are computed once at quantizer construction:

$$s_n^{\text{real}} = \text{headroom} \cdot \max_h |A_{\text{real}}[h, n]|, \tag{10}$$

$$s_n^{\text{imag}} = \text{headroom} \cdot \max_h |A_{\text{imag}}[h, n]|, \tag{11}$$

with headroom factor 1.05 as a safety margin against parameter drift. Scales are held fixed during training; learnable scales (LSQ-style) are left for future work.

*b. Per-tensor scales.* For the per-tensor ablation, $s = \text{headroom} \cdot \max_{h,n} |A[h, n]|$—one scale for the entire tensor of one type ($A_{\text{real}}$ or $A_{\text{imag}}$).

## IV. RESULTS

We report two task-level experiments on sCIFAR-1024: a full QAT sweep across initializations, bit-widths, and seeds (Section IV A); and an ablation freezing $\log \Delta$ at the Nyquist initialization to probe the source of conditioning erosion during training (Section IV B). Pre-training kernel-error sensitivity measurements that motivated the experimental design appear in Appendix C; the training procedure, diagnostic instrumentation, and experimental grid are detailed in Appendix B.

### A. Full QAT sweep on sCIFAR-1024

We train the model from Section III on sCIFAR-1024 (50,000 training examples, sequence length $L = 1024$, 10 classes) for 200 epochs across both initializations, six bit-width settings, and three seeds, yielding 36 cells. Per-channel deterministic quantization with Nyquist-clustered $\log \Delta$ initialization throughout.

*a. Test accuracy (Table I).* The headline observation is that the FouT/Skew-HiPPO gap is roughly constant ($\sim$ 3pp) across all bit-widths *including* fp32. The gap is not a quantization-specific effect—it is the fp32 baseline gap propagating through QAT.

*b. Quantization cost (Table II).* Each init incurs a $\sim$ 1.3–2.0pp accuracy drop going from fp32 down to $B = 2$. The drops are roughly equal between FouT and Skew-HiPPO, indicating that per-channel quantization absorbs most of the conditioning differential before it reaches the task.

*c. Conditioning trajectories (Table III).* The init-time vs. training-end $\kappa$ values reveal a striking property: *training collapses the $\kappa$ axis*. FouT begins at $\kappa(V) = 14$ but drifts to $\kappa_{\max}^{\text{global}} \approx 1.13 \times 10^{13}$ by end of training—a 12-order-of-magnitude blow-up. Skew-HiPPO begins at $\kappa(V) = 1.3 \times 10^{13}$ and ends at $1.74 \times 10^{13}$, essentially flat (already at the ceiling). The well-conditioned starting point of FouT does not persist; training dynamics push the eigenvalues toward configurations with near-degenerate discrete-time Vandermonde nodes regardless of init.

*d. Statistical significance.* Per-cell standard deviations across 3 seeds are $\leq 0.012$ for all entries in Table I. The fp32 FouT-vs-LegS gap of +2.9pp (s.e. $\approx 0.003$) is significant at $p < 0.001$ under a Welch $t$-test. All quantized gaps at fixed bit-width are similarly significant.

### B. Ablation: frozen $\log \Delta$

We test whether $\Delta$-drift during training is the mechanism that erodes FouT's conditioning advantage. We freeze $\log \Delta$ at its Nyquist initialization (eq. 8) and re-train under otherwise identical conditions. Six cells: FouT only, $B \in \{4, \text{fp32}\}$, three seeds. Implementation details in Appendix B 4.

The $\kappa_{\max}^{\text{global}}$ trajectory (Table IV) tells the story. With $\log \Delta$ frozen, $\kappa_{\max}^{\text{global}}$ reaches $1.01 \times 10^{13}$ by epoch 6 and remains essentially flat thereafter ($1.07 \times 10^{13}$ at epoch 74). For comparison, the unfrozen FouT cell ends training at $\kappa_{\max}^{\text{global}} = 1.13 \times 10^{13}$. The frozen $\kappa$ is within $\sim 10\%$ of the unfrozen $\kappa$ at the end of training—and is reached within the first six epochs.

The implication is direct: **$\Delta$-drift is not the load-bearing mechanism eroding FouT's conditioning**. With $\log \Delta$ fixed, $A_{\text{real}}$ and $A_{\text{imag}}$ drift alone is sufficient to push $\kappa$ to LegS magnitude within a small number of epochs. The Nyquist alignment of FouT is fragile to all SSM-parameter drift, not just to $\Delta$-drift. Preserving the well-conditioned init would require freezing $A_{\text{real}}$ and $A_{\text{imag}}$ as well—which would eliminate SSM learning entirely.

### C. Discussion

The convergence of differently-initialized diagonal SSMs to comparable $\kappa$ magnitudes despite starting 12 orders of magnitude apart demands a re-reading of HiPPO theory. HiPPO [1] establishes equivalence among polynomial-projection schemes *by construction*: different choices of measure $\mu(t)$ yield methods (LMU, leaky integrators, S4D-LegS) derivable as instantiations of the same projection operator. Our finding suggests a broader empirical equivalence: even initializations that are not HiPPO instantiations—FouT/S4D-Lin has no underlying polynomial measure, only a Fourier-aligned eigenvalue layout—converge under task-driven training to operating points with the same conditioning regime as HiPPO-derived inits. The HiPPO-optimal structure may be more accurately characterized as a *destination property* of diagonal-SSM training than as a property of the parameterization: gradient descent on a sufficiently expressive diagonal model recovers (or finds an analog of) the polynomial-projection-style operating point regardless of where it started. If this interpretation holds—a tighter test would compare the trained eigenvalue distributions directly, not just their condition numbers—then initialization choice is a choice of warm-start trajectory through a single attractor basin, not a choice between fundamentally different models.

Two empirical observations support this destination-equivalence reading. First, the Bauer–Fike bound (Appendix A) correctly predicts the qualitative ranking of initializations under quantization—FouT is less kernel-sensitive than Skew-HiPPO across all bit-widths—but

TABLE I. Test accuracy on sCIFAR-1024 (mean across 3 seeds; standard deviation in parentheses).

| init | $B=2$ | $B=3$ | $B=4$ | $B=6$ | $B=8$ | fp32 |
|---|---|---|---|---|---|---|
| FouT | 0.738 (.005) | 0.742 (.006) | 0.743 (.005) | 0.753 (.006) | 0.745 (.005) | 0.758 (.004) |
| Skew-HiPPO | 0.711 (.009) | 0.715 (.009) | 0.711 (.010) | 0.717 (.006) | 0.712 (.012) | 0.729 (.004) |
| gap | +2.7pp | +2.7pp | +3.1pp | +3.6pp | +3.2pp | +2.9pp |

TABLE II. Accuracy drop from fp32 baseline (percentage points).

| init | to $B=8$ | to $B=4$ | to $B=2$ |
|---|---|---|---|
| FouT | −1.3pp | −1.5pp | −2.0pp |
| Skew-HiPPO | −1.6pp | −1.7pp | −1.8pp |

TABLE III. Eigenvector matrix condition number $\kappa(V)$: init vs. end of training (across all layers, seed 0).

| init | $\kappa_{\text{init, layer 0}}$ | $\kappa_{\max}^{\text{global, final}}$ |
|---|---|---|
| FouT | 14.0 | $1.13\times10^{13}$ |
| Skew-HiPPO | $1.32\times10^{13}$ | $1.74\times10^{13}$ |

its quantitative magnitude does not translate to the per-channel task setting. The 12-order-of-magnitude $\kappa$ gap at initialization produces, at task level, a roughly constant $\sim$ 3pp accuracy gap that matches the fp32 baseline difference between the two inits. Per-channel quantization absorbs the $\kappa$-driven amplification before it reaches accuracy; init-time conditioning differences contribute to the trajectory but not to the destination.

Second, the frozen-$\Delta$ ablation tightens the conditioning-erosion finding: well-conditioned initialization does not survive training even with the timestep held fixed at the $\kappa$-minimizing value. The implication is that initialization-time conditioning is not predictive of deployment-time conditioning for diagonal SSMs, and that quantization-robustness analyses keyed to initialization-time $\kappa$ (including ours, originally) will overstate the practical relevance of the conditioning differential.

TABLE IV. $\kappa_{\max}^{\text{global}}$ trajectory, FouT $B=4$ seed 0.

| condition | $\kappa(V)$ at epoch $\sim 6$ | $\kappa(V)$ at training end |
|---|---|---|
| $\log\Delta$ frozen | $1.01\times10^{13}$ | $1.07\times10^{13}$ (epoch 74) |
| $\log\Delta$ unfrozen | $\sim 10^{10}$ | $1.13\times10^{13}$ (epoch $\sim 200$) |

## Appendix A: The Bauer–Fike bound for quantized diagonal SSMs

This appendix formalizes the conceptual statement in Section II A that perturbations to the diagonal parameters get amplified by $\kappa(V)$ to produce kernel error. We give the quantization model, the pseudospectral containment claim, and the Bauer–Fike kernel error bound. The bound is the formal anchor for the calibration discussion in Section C: it correctly predicts the ranking of initializations under quantization but is quantitatively loose by many orders of magnitude in the per-channel regime.

### 1. Quantization model

We model $B$-bit symmetric uniform quantization of a real-valued parameter $x$ as

$$Q_B(x) \;=\; \Delta_Q\cdot\mathrm{round}\Big(\tfrac{x}{\Delta_Q}\Big), \qquad \Delta_Q \;=\; \tfrac{s}{2^{B-1}-1}, \tag{A1}$$

where $s>0$ is a per-tensor or per-channel scale. The quantization error per element is bounded by $\Delta_Q/2$:

$$\big|Q_B(x)-x\big| \;\le\; \Delta_Q/2. \tag{A2}$$

Let $A_Q=\mathrm{diag}\big(Q_B(\mathrm{Re}\,\lambda_1)+i\,Q_B(\mathrm{Im}\,\lambda_1),\dots\big)$ denote the diagonal state matrix obtained by applying $Q_B$ elementwise to the real and imaginary parts of the eigenvalues of $A$ (Section II B). The corresponding perturbation $E:=A_Q-A$ satisfies $\|E\|_\infty\le\Delta_Q/2$.

For the formal results in this appendix, the choice of per-tensor vs. per-channel scale changes only the magnitude of $\Delta_Q$ (effectively replacing a single $\Delta_Q$ with a per-mode $\Delta_{Q,n}$); the bounds below apply per-mode when $\Delta_Q$ is interpreted accordingly. The empirically-relevant consequence of per-channel scaling is that the relevant $\Delta_Q$ is reduced for low-magnitude modes, which dominates the per-tensor scaling and is the proximate reason per-channel quantization closes the kernel-error gap predicted by Bauer–Fike (Section C).

### 2. Pseudospectral containment

Recall the standard pseudospectrum [6]: for $\epsilon>0$,

$$\Lambda_\epsilon(A) \;:=\; \big\{z\in\mathbb{C} \;:\; \|(zI-A)^{-1}\|\ge\epsilon^{-1}\big\}. \tag{A3}$$

Equivalently, $\Lambda_\epsilon(A)$ is the union of spectra of all matrices within distance $\epsilon$ of $A$ in operator norm.

**Proposition A.1** (Pseudospectral containment)**.** *Let $\Lambda^B(A)$ denote the spectrum of the quantized matrix $A_Q$ (Section A 1). Then*

$$\Lambda^B(A) \;\subset\; \Lambda_{\Delta_Q/2}(A). \tag{A4}$$

*Proof.* By (A2), $\|A_Q - A\|_\infty \le \Delta_Q/2$, and for diagonal matrices the operator norm coincides with the infinity norm. By the equivalent definition of the pseudospectrum as the union of spectra of perturbations of size $\le \epsilon$, the eigenvalues of $A_Q$ are contained in $\Lambda_{\Delta_Q/2}(A)$. $\square$

The practical content of Proposition A.1 is that QAT-induced eigenvalue perturbations live inside the continuous pseudospectrum of the unquantized operator, so all classical pseudospectral bounds apply to the quantized system. This in particular justifies measuring resolvent-norm quantities at the original (unquantized) $A$ to bound the quantized kernel error.

### 3. Bauer–Fike kernel error bound

Let $H(s) = C(sI - A)^{-1}B$ denote the SSM transfer function. Define

$$H_Q(s) \;=\; C(sI - A_Q)^{-1}B \tag{A5}$$

as the transfer function of the quantized system (with $B, C$ unchanged for the moment; the bound generalizes to quantized $B, C$ at the cost of additional $\|B\|, \|C\|$ factors).

**Theorem A.2** (Bauer–Fike kernel error). *Let $A$ be diagonalizable with $A = V\Lambda V^{-1}$, and let $A_Q$ be its $B$-bit quantization (Section A 1). For any $s \in \mathbb{C}$ with* $\mathrm{dist}(s, \Lambda_{\Delta_Q/2}(A)) > 0$,

$$\big\|H_Q(s) - H(s)\big\| \;\le\; \frac{c_1 \cdot (\Delta_Q/2)}{\mathrm{dist}\big(s,\, \Lambda_{\Delta_Q/2}(A)\big)} \cdot \kappa(V), \tag{A6}$$

*where $c_1 = \|B\|\,\|C\|$ and $\kappa(V)$ is the condition number of the diagonalizing eigenvector matrix.*

*Proof sketch.* Let $R_Q(s) := (sI - A_Q)^{-1}$ and $R(s) := (sI - A)^{-1}$. By the resolvent identity,

$$R_Q(s) - R(s) \;=\; R_Q(s)\,(A_Q - A)\,R(s).$$

Taking norms and using submultiplicativity,

$$\|R_Q(s) - R(s)\| \;\le\; \|A_Q - A\| \cdot \|R_Q(s)\| \cdot \|R(s)\|.$$

By Proposition A.1, the eigenvalues of $A_Q$ lie in $\Lambda_{\Delta_Q/2}(A)$, so by the pseudospectrum's definition (A3),

$$\|R_Q(s)\| \;\le\; \mathrm{dist}(s, \Lambda_{\Delta_Q/2}(A))^{-1}.$$

For the unperturbed resolvent, the standard Bauer–Fike theorem [5] gives $\|R(s)\| \le \kappa(V)/\,\mathrm{dist}(s, \Lambda(A))$, and since $\Lambda(A) \subset \Lambda_{\Delta_Q/2}(A)$, we have $\mathrm{dist}(s, \Lambda(A)) \ge \mathrm{dist}(s, \Lambda_{\Delta_Q/2}(A))$. Combining and using $\|A_Q - A\| \le \Delta_Q/2$ from (A2),

$$\|R_Q(s) - R(s)\| \;\le\; \frac{(\Delta_Q/2)\,\kappa(V)}{\mathrm{dist}(s, \Lambda_{\Delta_Q/2}(A))^2}.$$

The transfer-function form follows after multiplying by $\|C\|\|B\|$ and absorbing one factor of $\mathrm{dist}(s, \Lambda_{\Delta_Q/2}(A))^{-1}$ into the constant $c_1$ to recover (A6) in the form used in the main text. $\square$

Two remarks on (A6) bear on the empirical results.

First, the $\kappa(V)$ in (A6) is the condition number of the diagonalizing eigenvector matrix of $A$ *at the operating* $\Delta$. In the diagonal-S4D parameterization this is the discrete-time Vandermonde $V_{\ell,n} = \bar{\lambda}_n^{\ell}$ (Section II B); its conditioning is a function of both the continuous eigenvalues and $\Delta$, so the same initialization can produce dramatically different $\kappa(V)$ values at different timesteps.

Second, (A6) is derived under the worst-case bound $\|A_Q - A\| \le \Delta_Q/2$, which in the per-tensor scaling case is dominated by the largest-magnitude modes. Per-channel scaling effectively replaces a single $\Delta_Q$ with per-mode $\Delta_{Q,n}$, each scaled to its mode's dynamic range; the worst-case $\|A_Q - A\|$ in (A6) is then much smaller than what the per-tensor bound suggests, and the $\kappa(V)$-driven amplification is correspondingly reduced. This is the proximate mechanism by which per-channel quantization closes most of the predicted kernel-error gap; Section C reports the empirical magnitudes.

## Appendix B: Experimental setup

### 1. Model implementation

The diagonal state matrix is stored as two real tensors $A_{\mathrm{real}}, A_{\mathrm{imag}} \in \mathbb{R}^{H \times N/2}$, with

$$\lambda_{h,n} \;=\; -\exp(A_{\mathrm{real}}[h,n]) \;+\; i\,A_{\mathrm{imag}}[h,n]. \tag{B1}$$

Eigenvalues are reconstructed only inside the convolution kernel and only after quantization, so quantization noise enters in $(\alpha, \omega)$-space exactly as the Bauer–Fike analysis (Appendix A) assumes. The $-\exp(\cdot)$ on the real part enforces $\mathrm{Re}(\lambda) < 0$ without a clamp or projection step.

The input projection $B$ and output projection $C$ are stored as separate real/imaginary pairs $(B_{\mathrm{real}}, B_{\mathrm{imag}}, C_{\mathrm{real}}, C_{\mathrm{imag}}) \in \mathbb{R}^{H \times N/2}$ each. This sidesteps PyTorch's incomplete complex autograd and lets real-valued quantizers be plugged in for $B, C$ in future work without changes to the kernel. The per-channel log-timestep $\log \Delta \in \mathbb{R}^H$ and skip-connection scalar $D \in \mathbb{R}^H$ are real-valued.

### 2. Training procedure

We use AdamW with two parameter groups, following the canonical S4 / LRA convention:

- **SSM group:** $\{A_{\mathrm{real}}, A_{\mathrm{imag}}, \log\Delta, B_{\mathrm{real}}, B_{\mathrm{imag}}, C_{\mathrm{real}}, C_{\mathrm{imag}}\}$. Learning rate $0.1 \cdot \mathrm{lr}$, weight decay 0. The lower lr and absence of weight decay are canonical; weight decay on $B$ (initialized at all-ones in continuous time) would slowly collapse the kernel.

- **Other group:** encoder, layer-norm, classifier head, block linears, and skip-scalar $D$. Learning rate lr, weight decay 0.01.

Base learning rate $\mathrm{lr} = 4 \times 10^{-3}$ with cosine schedule and 10-epoch linear warmup. Gradient clipping at norm 1.0. We train for 200 epochs on sCIFAR-1024 with batch size 50 and a 10% held-out validation split.

### 3. Diagnostic measurements

To track the conditioning trajectory during training, each S4D layer is instrumented with a `diagnose()` method called once per epoch. It computes:

- **Per-channel runtime $\kappa$.** For each channel $h$, $\kappa(V_h)$ where $V_h$ is the discrete-time Vandermonde with eigenvalues $\bar{\lambda}_{h,n} = \exp(\lambda_{h,n} \cdot \exp(\log \Delta_h))$. We report median and max across channels in each layer, and the global max across all layers, denoted $\kappa_{\max}^{\text{global}}$.
- **Init-time $\kappa$.** The same computation evaluated at the canonical $\Delta = 2/N$, snapshotted at construction. This is the quantity that appears in the Bauer–Fike bound (Appendix A).

A separate **kernel-error sensitivity sweep** measures $\|H_Q(\omega) - H(\omega)\|/\|H(\omega)\|$ as a function of bit-width without retraining; results reported in Appendix C.

### 4. Frozen-$\Delta$ implementation

The frozen-$\Delta$ ablation marks $\log \Delta$ as non-trainable at construction so the optimizer skips it during the update step. All other SSM parameters $(A_{\text{real}}, A_{\text{imag}}, B, C, D)$ and non-SSM weights train normally under the standard schedule (Appendix B 2).

### 5. Experimental grid

The full sweep is the Cartesian product of initialization $\in \{\text{FouT}, \text{Skew-HiPPO}\}$, bit-width $\in \{2, 3, 4, 6, 8, \text{fp32}\}$, and seed $\in \{0, 1, 2\}$, yielding 36 cells. Two ablations supplement the main sweep: per-tensor quantization (12 cells) and frozen-$\Delta$ (6 cells, FouT only).

Each cell writes a row to a shared CSV with init choice, bit-width, seed, final validation and test accuracy, best epoch, $\kappa_{\text{init}}$ at layer 0, $\kappa_{\max}^{\text{global}}$ at end of training, and total wall-clock time.

## Appendix C: Phase 1: kernel-error sensitivity (pre-training)

To characterize the kernel-level perturbation gap predicted by Bauer–Fike before committing to task-level training, we measure the relative kernel error $\|H_Q(\omega) - H(\omega)\|/\|H(\omega)\|$ at initialization for FouT and Skew-HiPPO across bit-widths $B \in \{2, \ldots, 16\}$, under two quantization regimes (per-tensor vs. per-channel) and two rounding modes (deterministic vs. stochastic). The kernel-error sweep is performed without retraining: we instantiate the model, apply quantization at a given bit-width, and compute the kernel against the unquantized reference.

### 1. Per-tensor symmetric quantization

FouT relative kernel error recovers monotonically from 1.30 at $B = 2$ to 0.50 at $B = 8$ (Table V). Skew-HiPPO is pinned at $\sim 1.0$ across the entire range with no monotone improvement: per-tensor scales are dominated by the highest-magnitude modes, the residual perturbation is amplified by $\kappa(V) \approx 10^{13}$, and the kernel does not recover within the practical bit-width range. This is the regime where the Bauer–Fike analysis (eq. A6) has direct explanatory force.

TABLE V. Relative kernel error, per-tensor symmetric quantization.

| init | $B = 2$ | $B = 4$ | $B = 6$ | $B = 8$ |
|---|---|---|---|---|
| FouT | 1.30 | 0.91 | 0.65 | 0.50 |
| Skew-HiPPO | 1.00 | 1.00 | 1.00 | 1.00 |

### 2. Per-channel (per-mode) quantization

Both inits recover monotonically with $B$ and converge to the fp32 floor at high bit-widths (Table VI). FouT reaches 0.13 at $B = 8$ and $1.7 \times 10^{-3}$ at $B = 16$; Skew-HiPPO reaches 0.18 at $B = 8$ and $6.5 \times 10^{-3}$ at $B = 16$. The LegS/FouT ratio saturates at $\sim 3.9\times$ from $B = 12$ onward—*not* the $10^{12}\times$ ratio predicted by Bauer–Fike (eq. A6). Per-channel quantization absorbs the $\kappa$-driven amplification: per-mode perturbations are roughly uniform across the spectrum, and the bound's prefactor $\kappa(V)$ is multiplied by a much smaller per-mode quantization error.

TABLE VI. Relative kernel error, per-channel (per-mode) quantization.

| init | $B = 2$ | $B = 8$ | $B = 12$ | $B = 16$ |
|---|---|---|---|---|
| FouT | 1.19 | 0.13 | $8.0 \times 10^{-3}$ | $1.7 \times 10^{-3}$ |
| Skew-HiPPO | 0.99 | 0.18 | $2.4 \times 10^{-2}$ | $6.5 \times 10^{-3}$ |
| LegS/FouT | 0.83 | 1.4 | 3.0 | 3.8 |

### 3. Deterministic vs. stochastic rounding

At $B = 8$ per-mode, deterministic rounding gives FouT a substantial advantage over its stochastic counterpart

$(1.9 \times 10^{-2}$ vs. $7 \times 10^{-2})$, while Skew-HiPPO performs identically under both modes (Table VII). The mechanism is grid alignment: FouT's eigenvalues $\omega_n = \pi n$ align with the integer quantization grid at $\Delta = 2/N$ under round-to-nearest. Stochastic rounding adds $\sqrt{n}$-variance noise that destroys the alignment. Skew-HiPPO's irrational $\omega_n$ values do not align with any integer grid regardless of rounding mode.

TABLE VII. Relative kernel error at $B = 8$, per-mode quantization: deterministic vs. stochastic rounding.

| init | deterministic | stochastic |
|---|---|---|
| FouT | $1.9 \times 10^{-2}$ | $7.0 \times 10^{-2}$ |
| Skew-HiPPO | $1.4 \times 10^{-1}$ | $1.4 \times 10^{-1}$ |

#### 4. Timestep sensitivity of $\kappa(V)$

The FouT conditioning advantage is sharply $\Delta$-dependent (Table VIII). At $N = 64$, varying $\Delta$ off the Nyquist scale produces $\kappa$ values that match or exceed Skew-HiPPO's value within an order of magnitude of dt drift. The FouT advantage is not a robust property of the Fourier spectrum—it is a narrow phenomenon at $\Delta = 2/N$ where the discrete eigenvalues $\bar{\lambda}_n = e^{i\pi n \cdot 2/N}$ are uniformly distributed on the unit circle.

TABLE VIII. $\kappa(V)$ for FouT at varying $\Delta$, $N = 64$.

| $\Delta$ | $\kappa(V)$ |
|---|---|
| $2/N$ (Nyquist) | 14 |
| 0.01 | $1.4 \times 10^{13}$ |
| 0.001 | $5 \times 10^{13}$ |
| 0.1 | $5 \times 10^{13}$ |

#### 5. Implications for the main experiment

The Phase 1 measurements motivate three design choices used throughout Section IV A:

- **Per-channel quantization is the default** (Appendix C 2). Per-tensor's catastrophic LegS failure makes it uninformative as a task-level comparison—the model fails to train at all in that regime—so we restrict the main sweep to the per-channel setting where both inits are operational.
- **Deterministic rounding is the default** (Appendix C 3). FouT's grid-alignment advantage is washed out by stochastic noise.
- **Nyquist-clustered $\log \Delta$ initialization** (Appendix C 4). Only at $\Delta = 2/N$ does FouT achieve its $\kappa \approx 14$ baseline; even modest jitter off Nyquist puts it into the $10^{13}$ regime.

The $10^{12}\times$ kernel-error ratio predicted by Bauer–Fike is partially observable in per-tensor mode (Table V) but collapses to $\sim$ 3–4$\times$ in per-channel mode (Table VI). Section IV A tests whether this remaining 3–4$\times$ kernel-level gap translates to task accuracy; the quantization-cost table in the main text reports that it does not.


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